Onboard Operational Safety Filter for a Quadrotor in an Environment With Dynamic Obstacles

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***Abstract*—Quadrotors have been applied to a wide range of industrial applications in recent years. It is vital to ensure the safety of a quadrotor operated by a human pilot in a practical environment that usually involves dynamic obstacles. This article develops an onboard Operational Safety Filter (OSF) for a quadrotor in an environment with dynamic obstacles. The developed OSF integrates motion prediction of dynamic obstacles and an improved backup controller into the existing backup controller-based OSF framework. The motion prediction of dynamic obstacles aims to address the dynamic obstacles. The improved backup controller can provide enhanced operational freedom for a quadrotor with respect to the existing backup controllers. The developed onboard OSF is applied to quadrotors in simulation and in the real world to evaluate its effectiveness in addressing dynamic obstacles, improving operational freedom, and running on a real quadrotor.**

***Index Terms*—Control input regulation, dynamic obstacle avoidance, human pilot, operational safety filter (OSF), quadrotor.**

I. Introduction

QUADROTOR unmanned aerial vehicles (UAVs) have been applied to an increasing number of industrial applications (e.g., photogrammetry [1], rescue [2], surveillance [3], climate monitoring [4], industrial inspection [5], and logistics [6]) in recent years [7]. Quadrotors have also been empowered by machine learning techniques to outperform humans in areas such as quadrotor racing [8]. However, controlling a quadrotor manually in an environment with dynamic obstacles (e.g., a factory or a warehouse) is challenging for human pilots [9], [10]. In such an environment, safety is the primary concern, followed by efficiency and ergonomic considerations [11]. Human pilots are highly likely to fail to address dynamic obstacles in complex environments occasionally and lead to crashes. Providing safety guarantees (e.g., obstacle and crash avoidance) is vital for human-operated quadrotors in industrial applications [9], [12].

The safety of quadrotors in complex environments has been a hot research area in recent years. Scholars have done significant work on providing safe guarantees for quadrotors. Classic path planning approaches (e.g., A star and RRT) have been applied to quadrotors to find a feasible path and avoid obstacles [13]. Chen et al.[14] proposed an efficient risk-aware sampling-based local trajectory planner to achieve safe flights for quadrotors. Tordesillas et al.[15] established a fast and safe trajectory planner to plan a collision-free path based on the low computational ability of a quadrotor. Wang et al. [16] developed a safe visibility-guided perception-aware trajectory optimization approach for quadrotors. The proposed approach can address occlusion and collision in a complex environment simultaneously. A guided time-optimal model predictive control (MPC)-based approach to generate chasing trajectories for a quadrotor in a cluttered environment to chase an escaping target and avoid obstacles has been developed by Xi et al. [17]. Lu et al.[18] presented a parallel and gradient-free strategy for path planning of a quadrotor with a limited field of view in a complex environment. Zhang et al. [19] proposed to apply reinforcement learning and human demonstration to obtain a policy for the obstacle avoidance of a quadrotor. Nageli et al.[20] developed a method combining an artificial potential field and hard safe constraints encoded into MPC to ensure the safety of a quadrotor in an environment with dynamic obstacles. However, the abovementioned methods take over the control authority of a quadrotor and cannot provide a certain degree of operational freedom for a human pilot.

To regulate nominal control inputs for a quadrotor to keep the quadrotor safe, the concept of operational safety filter (OSF) has been proposed [21]. An OSF allows a human pilot to operate a quadrotor aggressively in a safe fashion. To achieve OSFs, MPC-based methods, such as model predictive safety certification [22], predictive safety filters [23], and predictive shielding [24], have been proposed to revise the unsafe nominal control inputs. Broad et al. [25] proposed a sampling and MPC-based approach to generate several possible safe trajectories at every time step, and choose the one closest to nominal control

Received 29 May 2024; accepted 28 November 2024. This work was supported in part by the National Science Foundation of China (NSFC) under Grant 12172109 and Grant 52405010, in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2022A1515011492 and Grant 2023A1515011010, and in part by the Shenzhen Science and Technology Innovation Program under Grant GXWD20231130150349002, Grant JCYJ20220531095605012, Grant KJZD20230923115210021, and Grant 29853MKCJ202300205. Paper no. TII-24-2642. *(Corresponding author: Hao Xiong.)*

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Digital Object Identifier 10.1109/TII.2024.3514167

inputs. Shih et al. [26] introduced a learning-based approach to develop a safe policy offline and deploy the safe policy online to determine whether the safe policy or a nominal policy should be applied. Control Barrier Functions (CBFs) [27] have been utilized to obtain OSFs in recent years also. CBF-based methods usually minimally revise nominal control inputs to ensure the safe of a system via solving a Quadratic Program (QP) problem. CBF-based methods have been developed for a wide variety of quadrotors [28], [29]. Wang et al. [30] applied CBFs and Gaussian Process to provide high probability safety guarantees for safe learning. However, due to the limited computational capacity of a quadrotor in the real world, the abovementioned methods can hardly be implemented onboard [21].

To address the limited computational capacity of a quadrotor, Singletary et al. [21] developed a vanilla backup controller and leveraged implicit CBFs based on only an onboard computer to provide formal guarantees of safe flight under the control constraints. Singletary et al. [31] also proposed a time-varying backup controller for a system of multiple quadrotors and presented a framework for increasing the operational freedom of a quadrotor. Wang et al. [32] developed a dual MPC approach that utilizes control Lyapunov functions and CBFs to achieve stability and safety for a swarm of quadrotors. However, the abovementioned methods cannot address dynamic obstacles.

This article extends the existing OSF [21] and develops an onboard OSF to regulate nominal control inputs (e.g., control inputs performed by a human pilot or determined by a flight controller) for a quadrotor in an environment with dynamic obstacles. An improved backup controller is proposed to increase the operational freedom of a quadrotor from the existing OSF also.

The major contributions of this article are as follows.

1) This article develops an onboard OSF to regulate the nominal control inputs of a quadrotor in an environment with dynamic obstacles to ensure the safety of the quadrotor. To the best knowledge of the authors, for the first time, the developed OSF can ensure the safety of a quadrotor in an environment with dynamic obstacles by regulating, rather than replacing, nominal control inputs based on an onboard computer.
2) This article proposes an improved backup controller that can lead to an enlarged implicit safe set than up-to-date backup controllers for an OSF. Thus, the developed onboard OSF can improve the operational freedom of a quadrotor compared to OSFs based on existing backup controllers while ensuring the safety of a quadrotor.
3) This article implements the developed onboard OSF to quadrotors in simulation and in the real world. Experiments are conducted to verify the effectiveness of the OSF in addressing dynamic obstacles, improving operational freedom, and running on a real quadrotor.

The novelties of this study lie in the development of an onboard OSF that ensures the safety of a quadrotor in an environment with dynamic obstacles by regulating nominal control inputs for the first time and the improvement of operational freedom of a quadrotor from the existing backup controllers on condition that the safety of the quadrotor is guaranteed by a backup controller. The novelties enhance the safety of a quadrotor in an environment with dynamic obstacles and the maneuvering performance of the quadrotor with a backup controller.

The rest of this article is organized as follows. Section II presents the preliminaries of this study. A novel OSF for a quadrotor in an environment with dynamic obstacles is proposed in Section III. In Section IV, the proposed OSF is compared to the existing OSFs in simulation and in the real world to evaluate its effectiveness in addressing dynamic obstacles, improving operational freedom, and running on a real quadrotor. Finally, Section V concludes this article.

II. Preliminaries

This section introduces the preliminaries of the article, including the simplified model of a quadrotor, problem statement, and an up-to-date OSF based on a time-varying backup controller.

A. Simplified Model of a Quadrotor

The simplified model of a quadrotor has been widely used by autopilots for quadrotors in practice [33]. According to a simplified model, the dynamics of a quadrotor $\mathbf{p} \in \mathbb{R}^3$ satisfies

$$\begin{cases} \dot{\mathbf{p}} = \mathbf{v} \\ \dot{\mathbf{v}} = g\mathbf{e}_3 + \frac{1}{m}F \begin{bmatrix} \cos(\phi)\sin(\theta) \\ -\sin(\phi) \\ \cos(\phi)\cos(\theta) \end{bmatrix} \end{cases} \tag{1}$$

where $\mathbf{v} \in \mathbb{R}^3$ is the translational velocity of the quadrotor. g is the gravity acceleration and $\mathbf{e}_3 = [0, 0, 1]^T$. m is the mass of the quadrotor. F is the total thrust of the quadrotor. ϕ is the roll angle of the quadrotor. θ is the pitch angle of the quadrotor. The control input of the quadrotor is $\mathbf{u} = [F, \phi, \theta]^T$. The state of the quadrotor is represented by $\mathbf{x} = [\mathbf{p}^T, \mathbf{v}^T]^T$. In practice, constraints for the control input should be considered, including $F \in [F_{\text{min}}, F_{\text{max}}]$, $\phi \in [\phi_{\text{min}}, \phi_{\text{max}}]$, and $\theta \in [\theta_{\text{min}}, \theta_{maax}]$.

B. Problem Statement

This article deals with an OSF for a quadrotor in an environment with dynamic obstacles. An OSF is required to regulate the nominal control inputs $\mathbf{u}_{\text{nominal}}$ of a quadrotor to maintain the state of the quadrotor within a safe set and perform the nominal control inputs partially. The safe set is defined as

$$\mathcal{S} = \{\mathbf{p} \in \mathbb{R}^3 | \mathbf{p} \in \mathcal{S}_{\text{pos}} \cap \mathbf{p} \in \mathcal{S}_{\text{obs}}\} \tag{2}$$

where

$$\mathcal{S}_{\text{pos}} = \{\mathbf{p} \in \mathbb{R}^3 | \mathbf{p}_{\text{min}} \leq \mathbf{p} \leq \mathbf{p}_{\text{max}}\} \tag{3}$$

$$\mathcal{S}_{\text{obs}} = \{\mathbf{p} \in \mathbb{R}^3 | ||\mathbf{p} - \mathbf{p}_{oi}|| > d_{\text{safe}}, i = 0, \ldots, n\} \tag{4}$$

where $\mathbf{p}_{\text{min}} = [p_{x\text{min}}, p_{y\text{min}}, p_{z\text{min}}]^T$ and $\mathbf{p}_{\text{max}} = [p_{x\text{max}}, p_{y\text{max}}, p_{z\text{max}}]^T$ denote the lower bound and upper bound of the position of the quadrotor, respectively [9]. $\mathbf{p}_{\text{min}}$ and $\mathbf{p}_{\text{max}}$ determine a box constraint on the position of the quadrotor. $\mathbf{p}_{oi}$ denotes the position of the ith obstacle. d_{safe} represents the safe distance between the quadrotor and an obstacle.

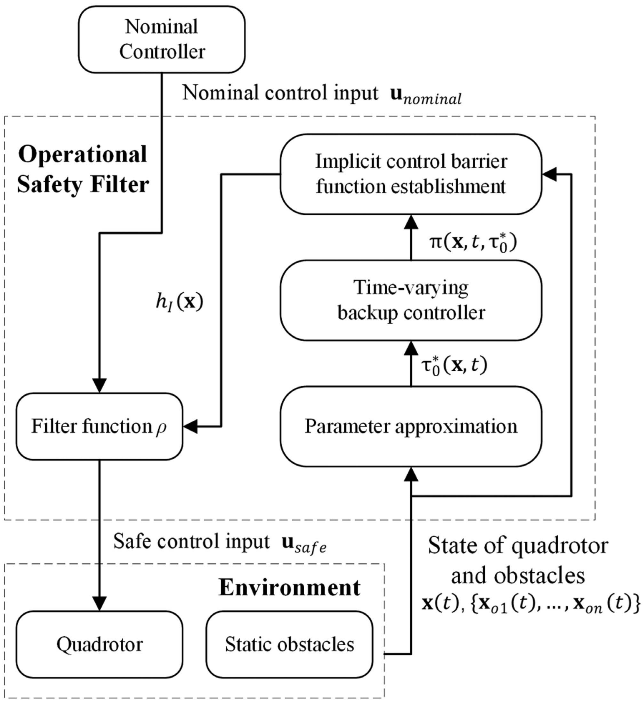


Fig. 1. Flow diagram of an OSF based on a time-varying backup controller [31].

C. OSF Based on a Time-Varying Backup Controller

Onboard OSFs have been developed for quadrotors in static environments in recent studies [21], [31]. The flow diagram of an OSF based on a time-varying backup controller [31] is presented in Fig. 1. The OSF recursively revises the nominal control input $\mathbf{u}_{\text{nominal}}$ determined by a nominal controller (e.g., a human pilot) and determines a safe control input $\mathbf{u}_{\text{safe}}$. Specifically, the OSF revises the nominal control input $\mathbf{u}_{\text{nominal}}$ according to the following steps. 1) Approximate the parameter τ_0^* used by a time-varying backup controller based on the state of the quadrotor $\mathbf{x}(t)$ and the state of obstacles $\{\mathbf{x}_{o1}(t), \ldots, \mathbf{x}_{on}(t)\}$; 2) Establish a time-varying backup controller, denoted as $\pi_{\text{TVBK}}(\mathbf{x}, t, \tau_0)$, according to the parameter τ_0^*; 3) Calculate the flow map of the system, denoted as $\Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t, \tau_0)$, according to the time-varying backup controller and establish an implicit CBF $h_I(\mathbf{x})$ based on the flow map; 4) Filter a nominal control input $\mathbf{u}_{\text{nominal}}$ according to a filter function based on the implicit CBF $h_I(\mathbf{x})$ and determine a safe control input $\mathbf{u}_{\text{safe}}$. The safe control input will render the safe set $\mathcal{S}$ forward invariant.

The OSF presented in [31] regards a quadrotor as a general system

$$\dot{\mathbf{x}} = f(\mathbf{x}) + g(\mathbf{x})\mathbf{u}. \tag{5}$$

The OSF is required to render a safe set $\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^n | h(\mathbf{x}) > 0\}$ forward invariant, where $h : \mathbb{R}^n \to \mathbb{R}$ is a continuous function. It is assumed that there exists a function $h_B(\mathbf{x}) : \mathbb{R}^n \to \mathbb{R}$ and the corresponding safe set $\mathcal{S}_B = \{\mathbf{x} \in \mathbb{R}^n | h_B(\mathbf{x}) > 0\} \subset \mathcal{S}$ is control invariant. $h_B : \mathbb{R}^n \to \mathbb{R}$ is a continuous function. The safe set $\mathcal{S}_B$ is called the backup set [34]. $\mathcal{S}_B$ corresponds to a controllable subset of the state space of the system. $\mathcal{S}_B$ can be significantly smaller than $\mathcal{S}$. The definition of control invariant is as follows.

Definition 1 (see[31]): Safe set $\mathcal{S}_B$ is control invariant if there exists a controller $\pi : \mathbb{R}^n \to U$ that renders $\mathcal{S}_B$ invariant.

Consider a controller

$$\pi_{\text{VBK}}(\mathbf{x}) = \mathbf{u}_{\text{bk}}(\mathbf{x}) \tag{6}$$

where $\pi_{\text{VBK}}(\mathbf{x})$ renders the backup set $\mathcal{S}_B$ forward invariant. $\mathbf{u}_{bk}(\mathbf{x})$ is a control input that can lead the state of a system to $\mathcal{S}_B$. $\pi_{\text{VBK}}(\mathbf{x})$ is a vanilla backup controller. For a quadrotor, $\mathbf{u}_{bk}(\mathbf{x})$ can be a velocity controller that stops the quadrotor [31].

One can define a corresponding implicit safe set [34]

$$\begin{aligned} \mathcal{S}_I(\pi_{\text{VBK}}) &= \{\mathbf{x} \in \mathbb{R}^n | \Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t) \in \mathcal{S} \\ &\qquad \cap \Phi_{\pi_{\text{VBK}}}(\mathbf{x}, T) \in \mathcal{S}_B, \forall t \in [0, T]\} \\ &= \{\mathbf{x} \in \mathbb{R}^n | h(\Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t)) \geq 0, \forall t \in [0, T] \\ &\qquad h_B(\Phi_{\pi_{\text{VBK}}}(\mathbf{x}, T)) \geq 0\} \end{aligned} \tag{7}$$

where

$$\Phi_{\pi_{\text{VBK}}}(\mathbf{x}(0), t) = \mathbf{x}(0) + \int_0^t (f(\mathbf{x}) + g(\mathbf{x})\pi_{\text{VBK}}(\mathbf{x}))dt \tag{8}$$

represents the flow map of the system with the initial state $\mathbf{x}(0)$. T represents the time horizon. The controller $\pi_{\text{VBK}}(\mathbf{x})$ that can lead to a nonempty $\mathcal{S}_I$ is a backup controller.

Definition 2 (see [31]): (Implicit CBF): For a system with the flow map $\Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t)$ and a backup controller π_{VBK}, one can define an implicit CBF h_I as

$$h_I(\mathbf{x}) = \min_{t \in [0,T]} \{h(\Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t)), h_B(\Phi_{\pi_{\text{VBK}}}(\mathbf{x}, T))\}. \tag{9}$$

The invariance of $\mathcal{S}$ can be achieved by applying a proper filter function $\mathbf{u}_{\text{safe}}(\mathbf{x}) = \rho(\mathbf{x}, h_I(\mathbf{x}), \mathbf{u}_{\text{nominal}}(\mathbf{x}))$ [34]. The filter function should satisfy the following conditions [31] - 1) A backup controller is applied when $\mathbf{x}$ reaches the boundary of $\mathcal{S}_I$; and 2) $\mathbf{u}_{\text{safe}}(\mathbf{x})$ is locally Lipschitz continuous. A filter function that satisfies both conditions can be [31]

$$\begin{aligned} \mathbf{u}_{\text{safe}}(\mathbf{x}) &= \rho(\mathbf{x}, h_I(\mathbf{x}), \mathbf{u}_{\text{nominal}}(\mathbf{x})) \\ &= \xi(\mathbf{x}, h_I(\mathbf{x}))\mathbf{u}_{\text{nominal}}(\mathbf{x}) \\ &\quad + (1 - \xi(\mathbf{x}, h_I(\mathbf{x})))\mathbf{u}_{bk}(\mathbf{x}) \end{aligned} \tag{10}$$

where $\xi(\mathbf{x}, h_I(\mathbf{x}))$ is an adjustment function for the weights of nominal and backup controllers given by

$$\xi(\mathbf{x}, h_I(\mathbf{x})) = 1 - \exp(-\beta \max(0, h_I(\mathbf{x}))) \tag{11}$$

where β is a tuning parameter. If $\xi(\mathbf{x}, h_I(\mathbf{x})) = 1$, the nominal controller is in domination. If the state of a system approaches the boundary of the safe set, the value of $\xi(\mathbf{x}, h_I(\mathbf{x}))$ and the weight of the nominal controller decrease. If the state of a system is close to the boundary of the safe set, the value of $\xi(\mathbf{x}, h_I(\mathbf{x}))$ will be zero and the safe controller will be in domination.

A time-varying backup controller can be designed as [31]

$$\pi_{\text{TVBK}}(\mathbf{x}, t) = \begin{cases} \mathbf{u}_{\text{nominal}}(\mathbf{x}) & t \leq T_M \\ \mathbf{u}_{\text{switch}}(\mathbf{x}, t) & T_M < t < T_M + \delta \\ \mathbf{u}_{bk}(\mathbf{x}) & t - \tau_0 \geq T_M + \delta \end{cases} \tag{12}$$

where $\mathbf{u}_{\text{switch}}(\mathbf{x}, t-\tau_0)$ represents a control input that continuously transitions from $\mathbf{u}_{\text{nominal}}(\mathbf{x})$ to $\mathbf{u}_{bk}(\mathbf{x})$ over a time interval δ. One should solve an optimization problem to achieve the optimal τ_0 as follows.

$$\begin{aligned}\tau_0^*(\mathbf{x}, t) := &\max_{\tau_0 \in [t-T, t]} \tau_0 \\ &\text{s.t. } \Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t+\tau, \tau_0) \in S \ \forall \tau \in [0, T] \\ &\quad\ \ \Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t+T, \tau_0) \in S_B \end{aligned} \tag{13}$$

where

$$\begin{aligned}\Phi_{\pi_{\text{TVBK}}}(\mathbf{x}(0), t, \tau_0) = \mathbf{x}(0) + \int_0^t f(\mathbf{x}) d\tau \\ + \int_0^t g(\mathbf{x}) \pi_{\text{TVBK}}(\mathbf{x}, \tau - \tau_0) d\tau.\end{aligned} \tag{14}$$

The time-varying backup controller can lead to an implicit safe set larger than the implicit safe set obtained by the vanilla backup controller according to [31].

III. OSF for a Quadrotor in an Environment With Dynamic Obstacles

In this section, an OSF is developed for a quadrotor in an environment with dynamic obstacles. Motion prediction of dynamic obstacles is integrated into the OSF to address dynamic obstacles. An improved backup controller is proposed to provide enhanced operational freedom for a nominal controller (e.g., a human pilot). The flow diagram of the OSF is presented in Fig. 2 and the improvements of the OSF are marked out.

An OSF for a quadrotor in an environment with dynamic obstacles recursively revises the nominal control input $\mathbf{u}_{\text{nominal}}$ determined by a nominal controller (e.g., a human pilot) and determines a safe control input $\mathbf{u}_{\text{safe}}$ also. Specifically, the OSF revises the nominal control input $\mathbf{u}_{\text{nominal}}$ according to the following steps. 1) Estimate the mean and covariance matrices on the state of obstacles and predict the future states of obstacles with associated uncertainty. The details of the estimation of the mean and covariance matrices on the state of obstacles are presented in Section III-A. 2) Approximate the parameters (λ_0^*, τ_0^*) used by an improved backup controller based on the state of the quadrotor $\mathbf{x}(t)$ and the state of obstacles $\{\mathbf{x}_{o1}(t), \ldots, \mathbf{x}_{on}(t)\}$. Establish an improved backup controller, denoted as $\pi_{\text{IBK}}(\mathbf{x}(0), t, \lambda_0, \tau_0)$, according to the parameter (λ_0^*, τ_0^*). The details of the improved backup controller are presented in Section III-C. 3) Calculate the flow map of the system, denoted as $\Phi_{\pi_{\text{IBK}}}(\mathbf{x}(0), t, \lambda_0, \tau_0)$, according to the improved backup controller and establish an implicit CBF $h_I(\mathbf{x})$ based on the flow map and the mean matrix on the state of obstacles. The details of the implicit CBF are presented in Section III-B. 4) Filter a nominal control input $\mathbf{u}_{\text{nominal}}$ according to a filter function based on the implicit CBF $h_I(\mathbf{x})$ and determine a safe control input $\mathbf{u}_{\text{safe}}$. The safe control input will render the safe set $\mathcal{S}$ forward invariant.

A. Motion Prediction of Dynamic Obstacles

To address dynamic obstacles, motion prediction of dynamic obstacles is integrated into the proposed OSF. Kalman filter, a widely used technique for recursively estimating the state of a system [35], [36], is used to estimate the state of dynamic obstacles, and then the path of dynamic obstacles is predicted. The uncertainty of the state of dynamic obstacles is modeled by a normal distribution according to [37]. For an obstacle, a Kalman filter can estimate the mean and covariance matrices on the state of the obstacle. The mean and covariance matrices can predict the future states of the obstacle with associated uncertainty. The predicted future states of dynamic obstacles provide a basis for the establishment of an implicit CBF that takes dynamic obstacles into account.

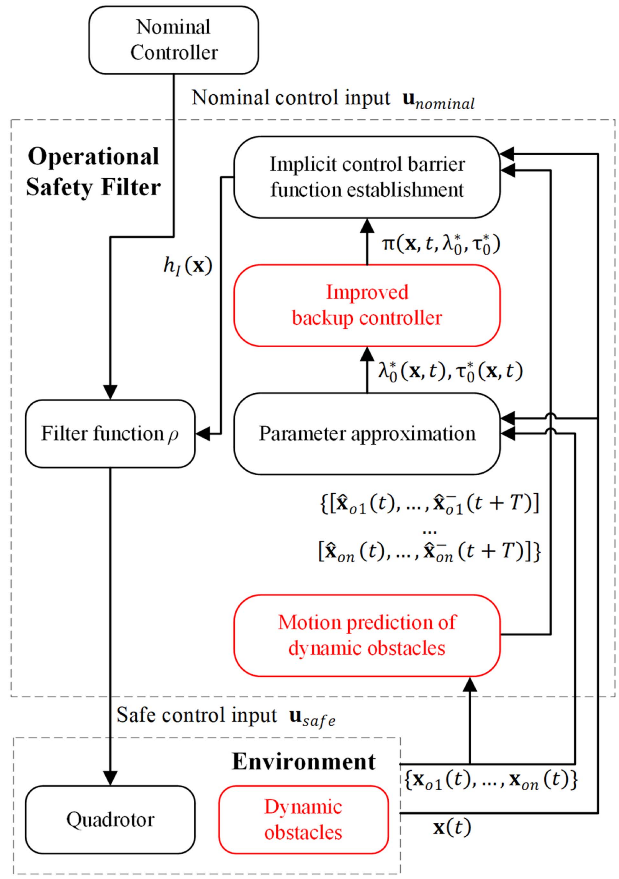


Fig. 2. Flow diagram of an OSF based on an improved backup controller for a quadrotor in an environment with dynamic obstacles.

Let $\mathbf{x}_{oi} = [\mathbf{p}_{oi}^T, \mathbf{v}_{oi}^T, \mathbf{a}_{oi}^T]^T$ represent the state of an obstacle. $\mathbf{p}_{oi}$, $\mathbf{v}_{oi}$, and $\mathbf{a}_{oi}$ are the position, velocity, and acceleration of the obstacle, respectively. A Kalman filter includes two recursive steps – the prediction step and the update step [35]. Let Δt represent the time step of the Kalman filter. In a prediction step, the priori mean state and the priori covariance matrix of the ith obstacle at time $k\Delta t$, denoted as $\hat{\mathbf{x}}_{oi}^-(k)$ and $\mathbf{P}_{oi}^-(k)$, respectively, are estimated according to [37]

$$\hat{\mathbf{x}}_{oi}^-(k) = \mathbf{A}\hat{\mathbf{x}}_{oi}(k-1) \tag{15}$$

$$\mathbf{P}_{oi}^-(k) = \mathbf{A}\mathbf{P}_{oi}(k-1)\mathbf{A}^T + \mathbf{Q} \tag{16}$$

where $\mathbf{A}$ is the state matrix. $\mathbf{Q}$ is the process noise covariance matrix. According to [37], the state matrix can be expressed as

$$\mathbf{A} = \begin{bmatrix} \mathbf{I} & \Delta t\mathbf{I} & \frac{\Delta t^2}{2}\mathbf{I} \\ \mathbf{0} & \mathbf{I} & \Delta t\mathbf{I} \\ \mathbf{0} & \mathbf{0} & \mathbf{I} \end{bmatrix}. \tag{17}$$

In an update step, the mean state and the covariance matrix of the ith obstacle at time $k\Delta t$, denoted as $\hat{\mathbf{x}}_{oi}(k)$ and $\mathbf{P}_{oi}(k)$, respectively, are calculated according to

$$\hat{\mathbf{x}}_{oi}(k) = \hat{\mathbf{x}}^-_{oi}(k) + \mathbf{K}_{oi}(k)(\mathbf{z}_{oi}(k) - \mathbf{H}\hat{\mathbf{x}}^-_{oi}(k)) \tag{18}$$

$$\mathbf{P}_{oi}(k) = (I - \mathbf{K}_{oi}(k)\mathbf{H})\mathbf{P}^-_{oi}(k) \tag{19}$$

where $\mathbf{H}$ is the observation matrix. The observation matrix can be expressed as [37]

$$\mathbf{H} = \mathbf{I}. \tag{20}$$

$\mathbf{z}(k)$ is the measurement of $\mathbf{x}(k)$. $\mathbf{K}(k)$ is the Kalman gain matrix

$$\mathbf{K}(k) = \mathbf{P}^-_{oi}(k)\mathbf{H}^T(\mathbf{H}\mathbf{P}^-_{oi}(k)\mathbf{H}^T + \mathbf{R})^{-1} \tag{21}$$

where $\mathbf{R}$ is the observation noise covariance matrix.

The priori mean state $\hat{\mathbf{x}}^-_{oi}$ and priori covariance matrix $\mathbf{P}^-_{oi}$ of the ith obstacle at a future step are predicted by looping in prediction steps according to (15) and (16). The mean state $\hat{\mathbf{x}}_{oi}$ of the ith obstacle at the current step is calculated according to (18). The covariance matrix $\mathbf{P}_{oi}$ of the ith obstacle at the current step is calculated according to (19). Then, $\hat{\mathbf{x}}_{oi}$ and $\hat{\mathbf{x}}^-_{oi}$, are used by the implicit CBF presented in Section III-B, as shown in Fig. 2.

B. Implicit CBF

To provide a safety guarantee for a quadrotor in an environment with dynamic obstacles, this section designs an implicit CBF, denoted as h_I, for the proposed OSF. The implicit CBF is based on the predicted motion of dynamic obstacles. According to the problem statement, a quadrotor is supposed to address a box constraint on the position and dynamic obstacles.

According to (9), $h(\mathbf{x})$ and $h_B(\mathbf{x})$ are designed to achieve an implicit CBF. $h(\mathbf{x})$ is designed according to the predefined safe set $\mathcal{S}$ in (2). $h_B(\mathbf{x})$ is designed based on the velocity of a quadrotor. $h(\mathbf{x})$ and $h_B(\mathbf{x})$ can be expressed as

$$h(\mathbf{x}) = \min(h_1(\mathbf{x}), h_2(\mathbf{x})) \tag{22}$$

$$h_B(\mathbf{x}) = \min(h_{B1}(\mathbf{x}), h_{B2}(\mathbf{x})) \tag{23}$$

where $h_1(\mathbf{x})$ is defined as

$$\begin{aligned} h_1(\mathbf{x}) = \min(&p_x - p_{x\min}, p_{x\max} - p_x, p_y - p_{y\min}, \\ &p_{y\max} - p_y, p_z - p_{z\min}, p_{z\max} - p_z) \end{aligned} \tag{24}$$

$h_1(\mathbf{x})$ is positive if the position of the quadrotor is within the set $\mathcal{S}_{\text{pos}}$. $h_2(\mathbf{x})$ is positive if the distance between the quadrotor and the ith obstacle is larger than the safe distance d_{safe}. $h_2(\mathbf{x})$ is defined as

$$h_2(\mathbf{x}) = \min_i(||\mathbf{p} - \mathbf{p}_{oi}|| - d_{\text{safe}}), i = 1, \ldots, n \tag{25}$$

where $\mathbf{p}_{oi}$ is the position of the ith obstacle. d_{safe} is the safe distance between the quadrotor and an obstacle.

With $h_1(\mathbf{x})$ and $h_2(\mathbf{x})$, the safe set for the quadrotor in (2) can be rewritten as

$$\mathcal{S} = \{\mathbf{x} \in \mathbb{R}^n | h_1(\mathbf{x}) \geq 0, h_2(\mathbf{x}) \geq 0\} \tag{26}$$

To define a backup set, $h_{B1}(\mathbf{x})$ and $h_{B2}(\mathbf{x})$ are defined as

$$h_{B1}(\mathbf{x}) = \epsilon - ||\mathbf{v}|| \tag{27}$$

$$h_{B2}(\mathbf{x}) = \min_i(h_{vo}(\mathbf{x}, \mathbf{x}_{oi})), i = 1, \ldots, n \tag{28}$$

where $h_{B1}(\mathbf{x})$ can slow the quadrotor down to a velocity of ϵ to guarantee that the quadrotor can stop when reaching the boundary of the safe set [21]. Inspired by the velocity obstacle approach [38], $h_{B2}(\mathbf{x})$ is designed to represent the minimum of a velocity obstacle function $h_{vo}(\mathbf{x}, \mathbf{x}_{oi})$

$$h_{vo}(\mathbf{x}, \mathbf{x}_{oi}) = \begin{cases} ||\mathbf{p} - \mathbf{p}_{oi}||^2 - d^2_{\text{safe}} & b \geq 0 \\ ((\mathbf{v} - \mathbf{v}_{oi})^T(\mathbf{p} - \mathbf{p}_{oi}))^2 & \\ -4||\mathbf{v} - \mathbf{v}_{oi}||^2(||\mathbf{p} - \mathbf{p}_{oi}||^2 & b < 0 \\ -d^2_{\text{safe}}) & \end{cases} \tag{29}$$

where $b = (\mathbf{v} - \mathbf{v}_{oi})^T(\mathbf{p} - \mathbf{p}_{oi})$. According to [38], $h_{vo}(\mathbf{x}, \mathbf{x}_{oi})$ is positive if the velocity of the quadrotor is outside of the velocity obstacle zone determined based on the velocity obstacle approach and negative otherwise. A quadrotor with a velocity within the velocity obstacle zone of an obstacle suggests that the quadrotor is highly possible to collide with the obstacle according to [38]. $h_{B2}(\mathbf{x})$ is positive if the quadrotor is outside of all velocity obstacle zones.

With $h_{B1}(\mathbf{x})$ and $h_{B2}(\mathbf{x})$, a backup set $\mathcal{S}_B$ can be defined as

$$\mathcal{S}_B = \{\mathbf{x} \in \mathbb{R}^n | h_{B1}(\mathbf{x}) \geq 0, h_{B2}(\mathbf{x}) \geq 0\}. \tag{30}$$

If the state of a quadrotor at every step can be controlled into the backup set $\mathcal{S}_B$ by a backup controller within a certain number of steps T, the state of the quadrotor $\mathbf{x}$ can be maintained within the implicit safe set $\mathcal{S}_I$ [31].

Based on the designed $h(\mathbf{x})$ and $h_B(\mathbf{x})$, an implicit CBF can be defined as

$$h_I(\mathbf{x}) = \min_{t\in[t_0,T]}\{h(\Phi_{\pi_{\text{IBK}}}(\mathbf{x}, t)), h_B(\Phi_{\pi_{\text{IBK}}}(\mathbf{x}, T))\} \tag{31}$$

Then, an implicit safe set $\mathcal{S}_I$ for a quadrotor in an environment with dynamic obstacles can be defined as

$$\mathcal{S}_I = \{\mathbf{x} \in \mathbb{R}^n | h_I(\mathbf{x}) \geq 0\} \tag{32}$$

C. Improved Backup Controller

This study improves the up-to-date time-varying backup controller proposed in [31] to increase operational freedom provided by an OSF. The improved backup controller includes a new parameter λ_0 in addition to the time-related parameter τ_0 included in the time-varying backup controller. This section proves that an optimized λ_0 can enlarge the implicit safe set and thus provide enhanced operational freedom for a quadrotor. The improved backup controller can be expressed as

$$\pi_{\text{IBK}}(\mathbf{x}, \lambda_0, t-\tau_0) = \mathbf{u}_{bk}(\mathbf{x}) + \mu(\lambda_0, t-\tau_0)(\mathbf{u}_{\text{nominal}}(\mathbf{x}) - \mathbf{u}_{bk}(\mathbf{x})) \tag{33}$$

where $\lambda_0 \in \mathbb{R}$ and $\tau_0 \in \mathbb{R}$. $\mathbf{u}_{\text{nominal}}$ is the nominal control input. μ is a continuous function defined as

$$\mu(\lambda, t) = \begin{cases} \lambda & t \leq T_M \\ f_s(\lambda, t) & T_M < t < T_M + \delta \\ 0 & t \geq T_M + \delta \end{cases} \tag{34}$$

where $f_s(t)$ represents a function that continuously decreases from λ to 0 over a time interval δ. It should be noticed that the time-varying backup controller proposed in [31] can be regarded as a special case of the improved backup controller defined in (33). One can set $\lambda_0 = 1$ for the improved backup controller to achieve the time-varying backup controller.

For a quadrotor with an initial state $\mathbf{x}(0) \in \mathbb{R}^n$ and an improved backup controller $\pi_{\text{IBK}}(\mathbf{x}, \lambda_0, t-\tau_0)$, the flow map of the quadrotor satisfies

$$\Phi_{\pi_{\text{IBK}}}(\mathbf{x}(0), t, \lambda_0, \tau_0) = \mathbf{x}(0) + \int_0^t f(\mathbf{x}) d\tau_0 + \int_0^t g(\mathbf{x}) \pi_{\text{IBK}}(\mathbf{x}, \lambda_0, \tau - \tau_0) d\tau. \tag{35}$$

The flow map can implicitly maintain the forward invariance of the following implicit safe set:

$$S_I = \{\mathbf{x} \in \mathbb{R}^n | h(\Phi_{\pi_{\text{IBK}}}(\mathbf{x}, t, \lambda_0, \tau_0)) \geq 0, \forall t \in [0, T] \; h_B(\Phi_{\pi_{\text{IBK}}}(\mathbf{x}, T, \lambda_0, \tau_0)) \geq 0\} \tag{36}$$

where τ_0 and λ_0 determine the shape of S_I. The improved backup controller can adjust τ_0 and λ_0 based on the current state of the quadrotor to enlarge S_I. Instead of adjusting τ_0 only according to (13), (λ_0, τ_0) is adjusted by solving the following optimization problem:

$$\begin{aligned} (\lambda_0^*(\mathbf{x}, t), \tau_0^*(\mathbf{x}, t)) = & \max_{\lambda_0 \in \mathbb{R}, \tau_0 \in [t-T, t]} \mu(\lambda_0, t-\tau_0) \\ & \text{s.t. } \Phi_{\pi_{\text{IBK}}}(\mathbf{x}, \tau, \lambda_0, \tau_0 - t) \in S, \forall \tau \in [0, T] \\ & \Phi_{\pi_{\text{IBK}}}(\mathbf{x}, T, \lambda_0, \tau_0 - t) \in S_B \\ & \mu(\lambda_0, t - \tau_0) \leq 1. \end{aligned} \tag{37}$$

By solving (λ_0^*, τ_0^*) and setting the (λ_0, τ_0) of the improved backup controller $\pi_{\text{IBK}}(\mathbf{x}, \lambda_0, t-\tau_0)$ to (λ_0^*, τ_0^*), the improved backup controller can maximize $\mu(\lambda_0, t-\tau_0)$. Then, enhanced operational freedom can be provided for the quadrotor, according to (33).

Inspired by [31], the authors find that (37) is an optimization problem that can be approximately solved efficiently. A method is proposed to approximately solve the optimization problem online and calculate (10) simultaneously. The method is presented in Algorithm 1. If a quadrotor is safe, λ_0 and τ_0 returned by Algorithm 1 will increase.

Algorithm 1: Method to Approximating (λ_0^*, τ_0^*) Online.

Input: current state $\mathbf{x}$, step t, time step Δt, previous solution $(\lambda_0^*(\mathbf{x}(t-\Delta t), t-\Delta t), \tau_0^*(\mathbf{x}(t-\Delta t), t-\Delta t)$, the improved backup controller π_{IBK}, $\Delta\lambda$, $\Delta\tau$.
Output: $(\lambda_0^*(\mathbf{x}, t), \tau_0^*(\mathbf{x}, t)))$
$\lambda_0 = \lambda_0^*(\mathbf{x}(t-\Delta t), t-\Delta t)$
$\tau_0 \leftarrow \tau_0^*(\mathbf{x}(t-\Delta t), t-\Delta t) + \Delta\tau - \Delta t$
if $\mu(\lambda_0, t-\tau_0) > 1$ **then**
  while $\mu(\lambda_0, t-\tau_0) > 1$ **do**
    $\lambda_0 \leftarrow \lambda_0 - \Delta\lambda$
  end while
else
  if $\mu(\lambda_0 + \Delta\lambda, t-\tau_0) < 1$ **then**
    $\lambda_0 \leftarrow \lambda_0 + \Delta\lambda$
  end if
end if
$\mathcal{T} \leftarrow \{\Phi_{\pi_{IBK}}(\mathbf{x}, t, \lambda_0, \tau_0) | 0 \leq t \leq T\}$
if $(\mathcal{T} \subseteq S \cap (\Phi_{\pi_{IBK}}(\mathbf{x}, T, \lambda_0, \tau_0)) \subseteq S_B)$ **then**
  $\lambda_0^*(\mathbf{x}, t) \leftarrow \lambda_0$
  $\tau_0^*(\mathbf{x}, t) \leftarrow \tau_0$
else
  $\lambda_0^*(\mathbf{x}, t) \leftarrow \lambda_0^*(\mathbf{x}(t-\Delta t), t-\Delta t)$
  $\tau_0^*(\mathbf{x}, t) \leftarrow \tau_0^*(\mathbf{x}(t-\Delta t), t-\Delta t) - \Delta t$
end if
return $(\lambda_0^*(\mathbf{x}, t), \tau_0^*(\mathbf{x}, t)))$

It is proved in the following paragraphs that updating (λ_0, τ_0) to $(\lambda_0^*(\mathbf{x}, t), \tau_0^*(\mathbf{x}, t))$ can enlarge the S_I obtained according to a vanilla backup controller [21] or a time-varying backup controller [31]. An enlarged S_I suggests enhanced operational freedom for a quadrotor.

Theorem 1: The implicit safe set S_I obtained by the improved backup controller $\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*)$, where (λ_0^*, τ_0^*) is determined based on (37), is larger than or equal to the S_I obtained by a vanilla backup controller $\pi_{\text{VBK}}(\mathbf{x})$ or a time-varying backup controller $\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)$, where $\tau_0'^*$ is determined by solving the optimization problem (13). The following equation holds:

$$S_I(\pi_{\text{VBK}}(\mathbf{x})) \subseteq S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)) \subseteq S_I(\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*)) \tag{38}$$

Proof: It should be noticed that $S_I(\pi_{\text{VBK}}(\mathbf{x})) \subseteq S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*))$ has been proven in [31]. One can assume that there exists an $\mathbf{x} \in S_I(\pi_{\text{VBK}}(\mathbf{x}))$ and $\mathbf{x} \notin S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*))$. According to the definition of $S_I(\pi_{\text{VBK}}(\mathbf{x}))$ in (7), $\mathbf{x} \in S_I(\pi_{\text{VBK}}(\mathbf{x}))$ implies that

$$\begin{cases} \Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t+\tau) \in \mathcal{S}, \forall \tau \in [0, T] \\ \Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t+T) \in \mathcal{S}_B \end{cases} \tag{39}$$

According to [31] and the definition of the time-varying backup controller in (12), if $\tau_0' = -T_M - \delta$, one has

$$\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0') = \mathbf{u}_{bk}(\mathbf{x}) = \pi_{\text{VBK}}(\mathbf{x}). \tag{40}$$

Then, according to the definition of the flow maps in (8) and (14), one has

$$\begin{cases}\Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t+\tau, \tau_0') = \Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t+\tau) \in \mathcal{S}, \forall t \in [0, T] \\ \Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t+T, \tau_0') = \Phi_{\pi_{\text{VBK}}}(\mathbf{x}, t+T) \in \mathcal{S}_B.\end{cases} \tag{41}$$

One can conclude that τ_0' is a feasible solution to the optimization problem (13). However, $\mathbf{x} \notin S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*))$ implies that for $\mathbf{x}$, the optimization problem (13) does not have a feasible solution, which is in contradiction with the conclusion that τ_0' is a feasible solution to the optimization problem (13). Thus, for all $\mathbf{x} \in S_I(\pi_{\text{VBK}}(\mathbf{x}))$, one can conclude that $\mathbf{x} \in S_I(\pi_{\text{VBK}}(\mathbf{x}, t-\tau_0'^*))$, suggesting that $S_I(\pi_{\text{VBK}}(\mathbf{x})) \subseteq S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*))$.

Then, one can focus on the proof of $S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)) \subseteq S_I(\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*))$. Assume that $S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)) \subseteq S_I(\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*))$ doesn't hold and there exists a state satisfying $\mathbf{x} \in S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0' *))$ and $\mathbf{x} \notin S_I(\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*))$. Similarly, according to the optimization problem (13), $\mathbf{x} \in S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0' *))$ implies that

$$\begin{cases}\Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t+\tau, \tau_0') \in \mathcal{S}, \forall t \in [0, T] \\ \Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t+T, \tau_0') \in \mathcal{S}_B\end{cases} \tag{42}$$

As shown in the paragraph following (34), $\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0)$ is a special case of $\pi_{\text{IBK}}(\mathbf{x}, \lambda_0, t-\tau_0)$ and one can set $\lambda_0 = 1$ and $\tau_0 = \tau_0'^*$ for $\pi_{\text{IBK}}(\mathbf{x}, \lambda_0, t-\tau_0)$ to achieve $\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)$ as

$$\pi_{\text{IBK}}(\mathbf{x}, 1, t-\tau_0'^*) = \pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*) \tag{43}$$

The flow map of the system satisfies

$$\Phi_{\pi_{\text{IBK}}}(\mathbf{x}(0), t, 1, \tau_0'^*) = \Phi_{\pi_{\text{TVBK}}}(\mathbf{x}, t, \tau_0'^*) \tag{44}$$

Thus, $(1, \tau_0'^*)$ is a feasible solution to the optimization problem (37). However, $\mathbf{x} \notin S_I(\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*))$ implies that for $\mathbf{x}$, the optimization problem (37) does not have a feasible solution, which is in contradiction with the conclusion that $(1, \tau_0'^*)$ is a feasible solution to the optimization problem (37). Thus, for all $\mathbf{x} \in S_I(\pi_{\text{TVBK}}(\mathbf{x}))$, one can conclude that $\mathbf{x} \in S_I(\pi_{\text{IBK}}(\mathbf{x}, t-\tau_0'^*))$, suggesting that $S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)) \subseteq S_I(\boldsymbol{\pi}_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*))$ must hold.

Then one can conclude that $S_I(\pi_{\text{VBK}}(\mathbf{x})) \subseteq S_I(\pi_{\text{TVBK}}(\mathbf{x}, t-\tau_0'^*)) \subseteq S_I(\pi_{\text{IBK}}(\mathbf{x}, \lambda_0^*, t-\tau_0^*))$. The theorem is proved.

Remark 1 ([21], [31]): Although there is likely to be a discontinuity in the input when setting a new (λ_0^*, τ_0^*), the resulting backup controller corresponding to the new (λ_0^*, τ_0^*) is still Lipschitz continuous according to the definition (33). According to [21, Theorem 1], if the backup controller is continuous and bounded, $\mathbf{u}_{\text{safe}}$ is locally Lipschitz continuous and the reset-induced discontinuities do not affect the safety guarantees.

IV. Experiments

To evaluate the effectiveness of the proposed OSF, experiments on a quadrotor are conducted in simulation and in an environment with dynamic obstacles in the real world. Constraints for the control inputs of a quadrotor in simulation and in the real world are listed in Tables I and II, respectively. The mass of the quadrotor is 1.15 kg. In the experiments, the safe distance between a quadrotor and an obstacle is set to $d_{\text{safe}} = 0.21$ m. Experiments in simulation are based on the MuJoCo simulator. A Proportional Derivative (PD) controller is used for the attitude control of a quadrotor. Experiments in the real world are based on a quadrotor prototype with a PX4 flight controller and an NVIDIA Jetson NX onboard computer. An OSF runs on the NVIDIA Jetson NX at a frequency of 50.00 Hz. The PX4 flight controller is used for the attitude control of the quadrotor prototype. A motion capture system is used to measure the position and velocity of the quadrotor and obstacles.

TABLE I
Constraints for the Control Input of a Quadrotor in Simulation

Parameters	Value	Parameters	Value
F_{max}	13.00 N	F_{min}	8.00 N
ϕ_{max}	0.35 rad	ϕ_{min}	-0.35 rad
θ_{max}	0.35 rad	θ_{min}	-0.35 rad

TABLE II
Constraints for the Control Input of a Quadrotor in the Real World

Parameters	Value	Parameters	Value
F_{max}	13.00 N	F_{min}	8.00 N
ϕ_{max}	0.15 rad	ϕ_{min}	-0.15 rad
θ_{max}	0.15 rad	θ_{min}	-0.15 rad

A. Implementation of Operational Safety Filters

Because of the limited computational capacity of a quadrotor [31], only OSFs based on a backup controller are investigated in experiments. The proposed OSF based on motion prediction of dynamic obstacles and an improved backup controller, referred to as *improved backup controller-based OSF*, is compared to an OSF based on a vanilla backup controller [21], referred to as *vanilla backup controller-based OSF*, and an OSF based on a time-varying backup controller [31], referred to as *time-varying backup controller-based OSF*. The vanilla backup controller, time-varying backup controller, and improved backup controller are defined in (6), (12), and (33), respectively. The $\mathbf{u}_{bk}$ in (6), (12), and (33) can expressed as $\mathbf{u}_{bk} = [F, \phi, \theta]^T$ and is determined based on a PD controller to achieve a zero velocity according to [31].

For all three OSFs, the predictive horizon is set to $T = 1.00$ s. The time step is set to $\Delta t = 0.02$ s. The parameters of an implicit CBF are $\epsilon = 0.10$ m/s. The parameter of a filter function is $\beta = 30.00$. For the proposed OSF and the OSF based on a time-varying backup controller, the parameters of a backup controller are set to $T_M = 0.10$ s and $\delta = 0.06$ s. For the proposed OSF, the parameters used for Algorithm 1 are $\Delta\lambda = 0.01$ and $\Delta\tau = 0.03$ s.

B. Operational Freedom Provided by OSFs

The effectiveness of the proposed improved backup controller-based OSF in improving the operational freedom of a quadrotor from the existing OSFs [21] and [31] is analyzed according to [31] in the third scenario. A quadrotor is supposed to track a target trajectory moving out of a safe region, as

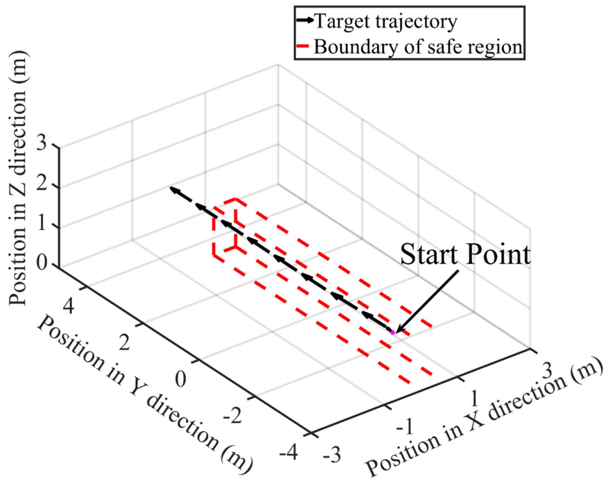
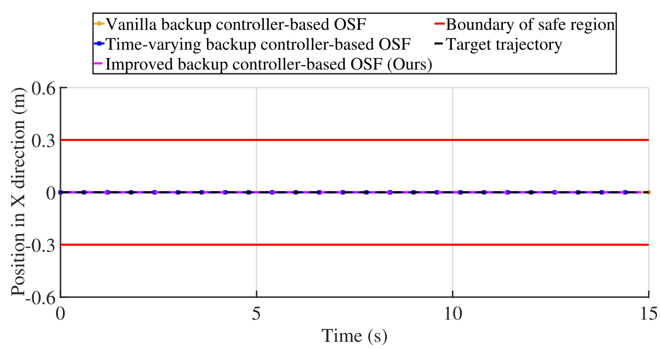
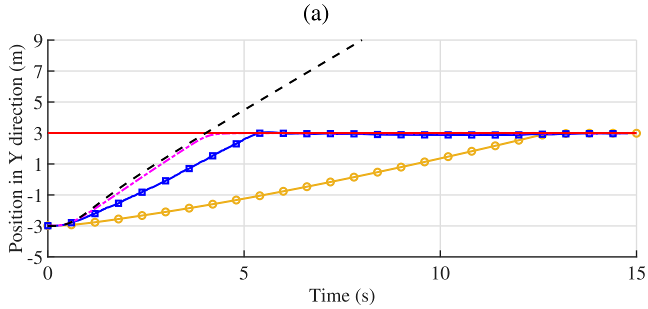


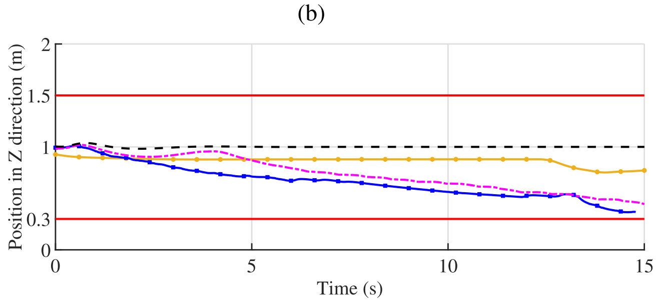
Fig. 3. Target trajectory and safe region of the third scenario.



(a)



(b)



(c)

Fig. 4. Position of quadrotors in (a) x direction, (b) y direction, and (c) z direction, when tracking a target trajectory moving out of a safe region.

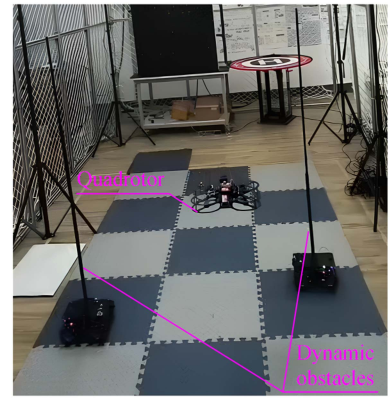


Fig. 5. Quadrotor in an environment with dynamic obstacles in the real world.

TABLE III
RANGE OF SAFE REGION OF THE THIRD SCENARIO

Parameters	Value	Parameters	Value
p_{xmax}	0.30 m	p_{xmin}	-0.30 m
p_{ymax}	3.00 m	p_{ymin}	-8.00 m
p_{zmax}	1.50 m	p_{zmin}	0.30 m

shown in Fig. 3. The target trajectory is designed to evaluate operational freedom provided by OSFs while ensuring safety, according to [31]. To provide large operational freedom for a quadrotor, an OSF should enable the quadrotor to be as close to the target trajectory as possible if the target trajectory is within the safe region. The target trajectory can be expressed as $\mathbf{p} = [0.00, -3.00 + 1.50t, 1.00]^T$ m. The safe region is defined in Table III. The initial state of the quadrotor is

$$\mathbf{x}(0) = [0.00\,\mathrm{m}, -3.00\,\mathrm{m}, 1.00\,\mathrm{m}$$
$$0.00\,\mathrm{m/s}, 0.00\,\mathrm{m/s}, 0.00\,\mathrm{m/s}]^T. \tag{45}$$

The position of quadrotors is plotted in Fig. 4. Fig. 4(b) shows that the improved backup controller-based OSF enables the quadrotor to be closest to the target trajectory from 0 to 5 s. According to [31], this phenomenon suggests that the improved backup controller-based OSF can provide enhanced operational freedom for a quadrotor than the existing OSFs.

C. Application of the Proposed OSF to a Human-Operated Quadrotor in the Real World

To evaluate the effectiveness of the proposed improved backup controller-based OSF in running on a real quadrotor, the proposed OSF is implemented to a real quadrotor to address two dynamic obstacles in the fourth scenario, as shown in Fig. 5. The quadrotor is controlled by a human pilot to pass through dynamic obstacles within a safe region. The target trajectories of the two dynamic obstacles are trajectories with periodic y element and constant x element. The target trajectories of the two cylindrical dynamic obstacles are designed to prevent the quadrotor from passing through. The safe region is defined in Table IV. A control input of the human pilot is $\mathbf{u}_{\text{nominal}} = \mathbf{u}_{\text{human}} = [u_F, u_\phi, u_\theta]^T$.

TABLE IV
RANGE OF SAFE REGION IN THE REAL WORLD

Parameters	Value	Parameters	Value
p_{xmax}	1.50 m	p_{xmin}	-1.00 m
p_{ymax}	0.50 m	p_{ymin}	-0.30 m
p_{zmax}	0.90 m	p_{zmin}	0.30 m

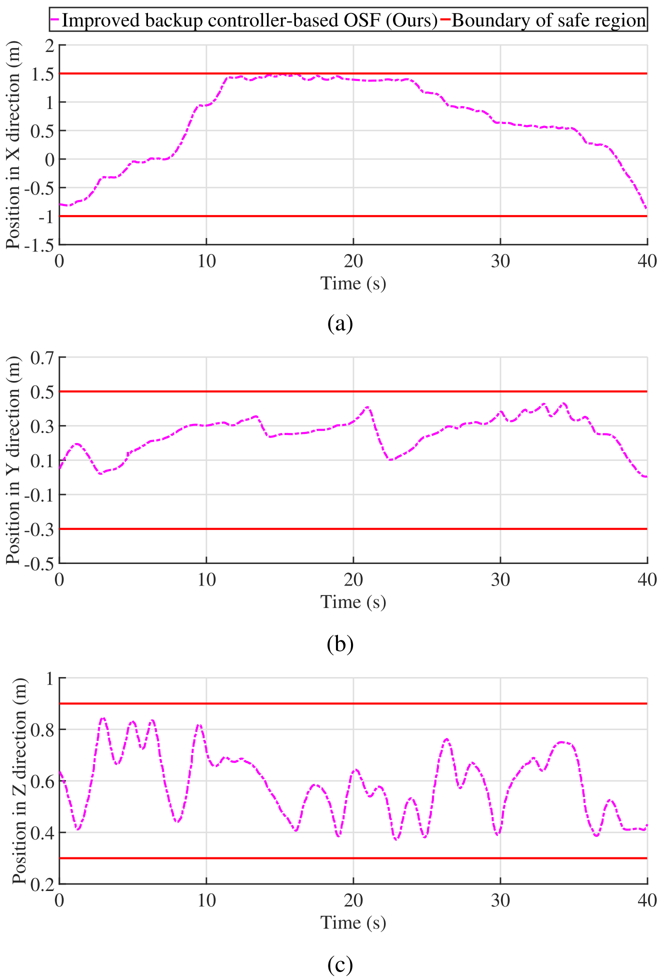


Fig. 6. Position of the quadrotor in the real world in (a) *x* direction, (b) *y* direction, and (c) *z* direction.

The position of the quadrotor is shown in Fig. 6. The nominal and safe control inputs of the improved backup controller-based OSF are shown in Fig. 7. According to Fig. 6, with the proposed improved backup controller-based OSF, the quadrotor controlled by a human pilot can pass through dynamic obstacles safely. Based on the estimated position and velocity of dynamic obstacles, the improved backup controller-based OSF switches between nominal control input and safe control input frequently to maintain the quadrotor within the safe region, according to Fig. 7. Figs. 6(a) and 7 show that the quadrotor stops at the boundary of the safe region even if the human pilot tries to control the quadrotor to fly out of the safe region. The effectiveness of the improved backup controller-based OSF in addressing dynamic obstacles, maintaining a quadrotor in a safe region, and running on a real quadrotor is verified in the scenario.

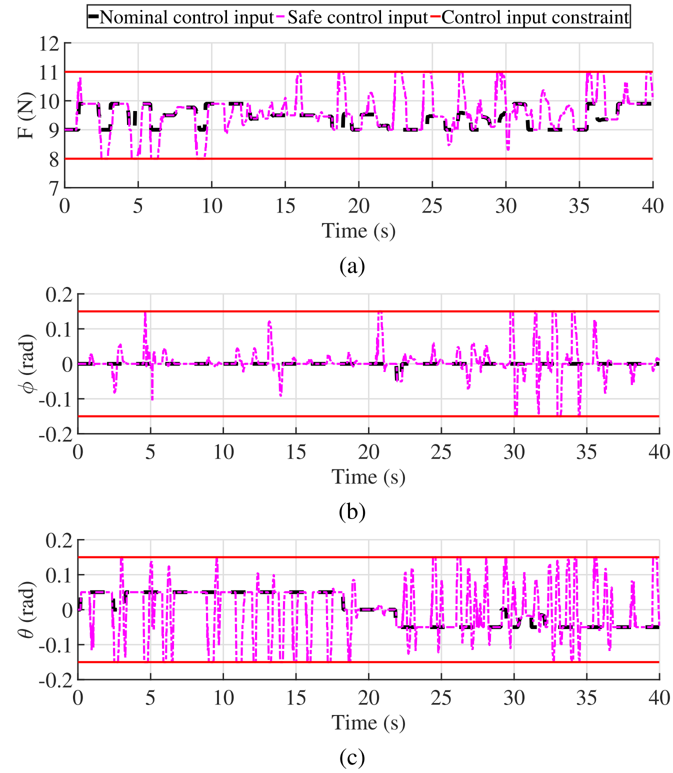


Fig. 7. Nominal and safe control inputs of the improved backup controller-based OSF in the real world for (a) thrust F, (b) roll ϕ, and (c) pitch θ.

V. CONCLUSION

This article developed an onboard OSF for a quadrotor in an environment with dynamic obstacles based on motion prediction of dynamic obstacles and an improved backup controller. The developed improved backup controller-based OSF can regulate nominal control inputs for a quadrotor (e.g., provided by a human pilot) to make the quadrotor avoid dynamic obstacles and maintain within a predefined safe region. The developed OSF has been applied to quadrotors in simulation and in the real world. The test of tracking a target trajectory moving out of a safe region shows that a quadrotor with the developed OSF can be much closer to the target trajectory than a quadrotor with OSFs based on up-to-date backup controllers. This suggests that the developed OSF can provide enhanced operational freedom for a quadrotor. The developed OSF has been deployed on the NVIDIA Jetson NX of a real quadrotor to address dynamic obstacles successfully, running at a frequency of 50 Hz. The performance of the developed OSF in addressing dynamic obstacles, improving operational freedom, and running on a real quadrotor has been verified.

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